Deterministic Particle Propagation for Next-Layer Cumulant Channels

Summary The proposed scheme is **Deterministic Nyström-Tail Compression (dNTC)**. For each of the k alive sources at a pre-activation layer, it carries a landmark cross-matrix $C_s^{(\mathrm{land})} \in \mathbb{R}^{m\times n}$ with $m = 120$ fixed landmark coordinates, a diagonal correction vector $\delta_s \in \mathbb{R}^n$, a small core tensor $\mathcal{T}_s^{(\mathrm{core})} \in \mathbb{R}^{m\times m\times m}$ for third-order contributions, and a length-n vector $\kappa_{s,4}^{(\mathrm{diag})}$ for fourth-order diagonals. No full $n \times n$ dense legs or explicit κ_3 tensor are stored; the per-source memory is $O(mn)$ instead of $O(n^2)$. The per-layer cost is dominated by updating the landmark matrices and forming an approximate full covariance via a low-rank plus diagonal representation, consuming at most 15 square products (measured in n^3-equivalent multiply-adds), which respects the ≤ 125u budget. The one falsifier is a **free-run channel-wise error gate**: propagate the dNTC state from layer 1, extract per-neuron variance, $d_3 = \kappa_3(z_i, z_i, z_i)$, κ_4 diagonal, and off-diagonal Y_{21} at layers 7/11/14/15, compare them against the 60M-sample teacher arrays using relative errors, and kill the scheme if any channel error exceeds $1.5\times$ the chain's reference error on that channel at two or more deep layers or if the error growth with depth is steeper than the chain's.

Problem Definition: Required Channels and Chain Constraints

Target Channels: Per-Neuron Variance, Third Cumulant Diagonal, Kurtosis Diagonal, and Their Feeder Covariances

At every pre-activation layer (l) the moment-propagation engine must deliver four groups of statistics for the vector (\mathbf{z}\in\mathbb{R}^n) ((n=1024)). These statistics feed the computation of the post-ReLU mean and variance, and ultimately the network's output mean.

1. **Per-neuron variance** – the diagonal entries of the covariance matrix
   [C_{ii} = \kappa_2(z_i,z_i)=\operatorname{Var}(z_i),\qquad i=1,\dots,n.]
2. **Per-neuron third cumulant** – the diagonal of the third cumulant tensor with identical arguments
   [d_3(i)=\kappa_3(z_i,z_i,z_i)=\mathbb{E}[(z_i-\mu_i)^3].]
3. **Per-neuron fourth cumulant diagonal**
   [K_{4d}(i)=\kappa_4(z_i,z_i,z_i,z_i).]
4. **Covariance entries that feed the above diagonals** – the off-diagonal structure of the second and third cumulants directly enters the formulas for (d_3(i)) and (K_{4d}(i)). The pre-activation is a linear combination of (k) independent source vectors from the previous layer, [\mathbf{z}=\sum_{s=1}^{k} W_s,\mathbf{a}_s,] where the (\mathbf{a}_s) are

post-activation vectors and the (W_s) are weight sub-matrices. The required feeder quantities are:

- the full covariance matrix (C_{ij}=\kappa_2(z_i,z_j)) for all (i,j);
- the matrix (Y_{21}) defined by [(Y_{21})_{ij} = \kappa_3(z_i,z_i,z_j),\qquad i,j=1,\dots,n,] i.e. the third cumulant tensor with two identical indices and one distinct index.

Among these, it is (Y_{21}) (referred to in the chain diagnostics as D21) that dominates the error of the current mixed-rank scheme. Any candidate deterministic scheme must reproduce all four channel groups with at least the fidelity of the chain’s own errors on the same layers.

The Chain’s Mixed-Rank Source Representation

At layer (l) the number of alive sources is (k=l-1) (the input itself is the sole source at (l=1)). By depth, (k) grows as:

Layer	Number of sources ((k))
7	6
11	10
14	13
15	14

Each source (s) is characterised by a dense leg: two (n\times n) matrices, (P^{(s)}) (the source’s contribution to the pre-activation covariance) and (A^{(s)}) (the linear map from the previous layer’s post-activation mean and variance to the source’s contribution). Both are of full rank (n=1024) for a newly created source. Storing and operating on all sources densely would require (O(k,n^2)\approx k\cdot 10^6) numbers, far beyond the available budget.

To limit cost, the chain compresses older sources according to an age schedule:

- **Ages 1–4** (recent sources): kept dense, (1024) directions each;
- **Ages 5–7**: represented in a shared basis of rank (384);
- **Ages (\ge 8)**: represented in a nested sub-basis of rank (224).

The compression is applied to the per-source matrices (P^{(s)}) and (A^{(s)}) individually, lowering the number of stored parameters. The chain’s cost is near the budget ceiling of **⩽ 125 u** (one “u” ≈ one (1024\times 1024) matrix product), translating to roughly 15 square products per layer.

Why the Hadamard Products for (Y_{21}) Remain Full Rank

The matrix (Y_{21}) can be written as a sum over sources of terms that involve the Hadamard (element-wise) product of (P^{(s)}) and (A^{(s)}). For a source with dense legs (P^{(s)}) and (A^{(s)}), its contribution takes the form [Y_{21}^{(s)} = \bigl(A^{(s)}\odot P^{(s)}\bigr),M^{(s)} ;+; \text{lower-order corrections},] where (\odot) denotes the Hadamard product and (M^{(s)}) incorporates the source’s mean and the weight geometry. Crucially, the Hadamard product of

two matrices of numerical rank (r) (even when each factor is well represented by a rank-(r) factorisation) typically does **not** inherit low-rank structure—its rank can reach (r^2) and is frequently full ((n)). In the chain’ s compressed representation, older sources are stored as (A^{(s)} = U_A^{(s)}V_A^{(s)\top}) and (P^{(s)} = U_P^{(s)}V_P^{(s)\top}) with (U,V\in\mathbb{R}^{n\times r}) and (r\in{384,224}). The Hadamard product becomes [A^{(s)}\odot P^{(s)} = \bigl(U_A^{(s)}V_A^{(s)\top}\bigr) ;\odot; \bigl(U_P^{(s)}V_P^{(s)\top}\bigr),] which does not factorise compactly; evaluating it inevitably requires forming and multiplying full (n\times n) matrices. Consequently, approximating (Y_{21}) by keeping only the compressed versions of the oldest sources introduces a structural error that cannot be eliminated by modest rank increases. Chain observations confirm this: discarding sources older than 10 layers while retaining the mixed-rank scheme still yields a **16% off-diagonal Frobenius error in D21 at layer 15**. Any alternative deterministic scheme must therefore avoid the full (n\times n) Hadamard product while still delivering the off-diagonal third-cumulant information required by (Y_{21}) and, through it, the per-neuron (d_3) and (K_{4d}) channels.

Baseline Errors and Falsification Protocol

The current chain’ s off-diagonal relative Frobenius error in (Y_{21}) (D21) against the teacher Monte Carlo arrays serves as the reference:

Layer	Chain D21 off-diagonal error
7	4.6%
11	5.7%
14	6.2%
15	6.6%

For the **Stage 1 gate** we will measure the same metric on the proposed scheme (under teacher-forcing or free-run, as appropriate) along with the relative errors for variance diagonal, (d_3) diagonal, (K_{4d}) diagonal, and the off-diagonal part of (Y_{21}). The scheme is killed immediately if **any** channel’ s error exceeds **1.5× the chain’ s error at two or more deep layers**, or if the error grows with depth faster than the chain’ s trend.

The **Stage 2 decisive test** integrates the scheme into the chain and measures the final-layer mean MSE against a (10^9)-sample truth. The target score is (\text{MSE}\times\text{cost} \le 0.44\times(\text{chain})), which corresponds to:

- MSE ⩽ 1.07× chain at ⩽ 102 u,
- MSE ⩽ 0.95× chain at 115 u,
- MSE ⩽ 0.84× chain at 125 u.

A prototype qualifies as a plausible path if it reaches a score ⩽ 0.8×–chain (e.g., MSE ⩽ 1.6× at 125 u) and the specific missing term that would close the remaining gap can be named. Below that, the approach is considered dead.

1. Particle-Ensemble Source Representation for Deep Pre-Activation Channels

1.1 Motivation

The chain' s current mixed-rank representation keeps per-source dense P and A matrices ($n \times n, n = 1024$, full rank) for sources up to age 4, and compresses older sources into shared or nested bases. This collapses at the off-diagonal third cumulant $Y_{21} = \kappa_3(z_i, z_i, z_j)$, because the elementwise Hadamard products needed for Y_{21} are not low-rank in the compressed bases. Retaining all dense legs would cost $O(kn^2)$ memory and $O(kn^3)$ compute, which is infeasible as the number k of alive sources grows with depth (from 6 at L7 to 14 at L15). Removing sources older than 10 layers still leaves a 16% relative error in Y_{21} at L15. We need a scheme that computes the next-layer means and all required higher-order channels—per-neuron variance, diagonal skewness $d_3 = \kappa_3(z_i, z_i, z_i)$, diagonal kurtosis K_{4d}, and the off-diagonal Y_{21} entries—without carrying the full κ_3 tensor or the per-source dense legs.

1.2 Core Idea

Replace each alive source by a fixed set of M deterministic particle vectors in $\mathbb{R}^n$. The particles are chosen at source birth so that the empirical ensemble matches the source' s marginal moments (mean, variance, skewness, kurtosis) up to fourth order, and respects the isotropic covariance structure. Under affine transformations (weight matrices) the particles evolve exactly; elementwise ReLU is applied particle-wise. The pre-activation at layer ℓ is the sum of contributions from all sources, giving a total cloud of $K = k\,M$ particles. All required inter-source channels—mean, covariance, d_3, K_{4d}, and Y_{21}—are extracted directly from this cloud as empirical cumulants. No $n \times n$ per-source dense legs and no full κ_3 tensor are ever stored or manipulated.

1.3 State and Cost

State carried per layer. For each of the k alive sources we keep an $M \times n$ matrix $V^{(s)}$ whose rows are the M particles of that source. Total storage is kMn floats. With $M = 32$ and $k \le 14$, this is at most $14 \times 32 \times 1024 \approx 459 \times 10^3$ numbers, negligible relative to the teacher arrays.

Cost in n^3 products (one unit $\equiv$ one $n \times n$ product, i.e. n^3 multiply–adds). The heaviest step is computing Y_{21} from the K-particle cloud. For each ordered off-diagonal pair (i, j), the empirical third joint cumulant costs $O(K)$. Filling the $\approx n^2/2$ independent entries therefore takes $O(Kn^2)$ operations. In units of n^3 products this is $Kn^2/n^3 = K/n = kM/n$. With $k = 14$, $M = 32$, $n = 1024$, we obtain $448/1024 \approx 0.44$ units. Adding the linear propagation of particles through W costs $2kMn^2$ multiply–adds, i.e. $2kM/n \approx 0.88$ units. All other cumulants (covariance, d_3, K_{4d}) require lower order work. The total per layer stays below 2 units, comfortably inside the budget of ≤ 15 square products per layer and well under the 125 u ceiling.

1.4 Deterministic Initialization of Particle Ensembles

At birth, a source is an isotropic random vector $x \in \mathbb{R}^n$ with zero mean, covariance $\sigma_s^2 I_n$, and known univariate skewness γ_s and excess kurtosis τ_s (initially Gaussian for depth-gated sources, later a pushforward through ReLU). We construct an M-particle ensemble that satisfies the following exactly:

- The empirical covariance equals $\sigma_s^2 I_n$.
- All third-order mixed moments $\mathbb{E}[x_i x_j x_k]$ vanish, so the off-diagonal Y_{21} at birth is exactly zero.
- For every coordinate direction, the empirical first four moments match the target marginal distribution: mean 0, variance σ_s^2, skewness γ_s, kurtosis $3 + \tau_s$.

The construction uses a spherical 3-design. A spherical t-design on $\mathbb{S}^{n-1}$ is a finite set of M_0 unit vectors $\{u_m\}$ such that for every polynomial of degree $\le t$, the arithmetic mean over the design equals the integral over the uniform measure on the sphere. For $t = 3$, designs with $M_0 = 2n$ (the cross-polytope, i.e. the coordinate axes $\pm e_i$) exist for any n. Augmenting with additional points from the orbit of a suitable reflection group gives a 3-design of size M_0 as low as $2n + 2\lfloor n/2 \rfloor$ or similar; for $n = 1024$ a 3-design with $M_0 = 32$ can be obtained numerically (e.g. by minimizing the spherical cap discrepancy). The key property is that empirical moments of odd degree vanish, guaranteeing isotropy up to order 3.

We then assign two radial levels to the direction vectors. Let the directions be paired into antipodal sets. For each antipodal pair $\pm u$, we choose two radii r_1, r_2 and weights w_1, w_2 (with $w_1 + w_2 = 1$) such that the average of r^2 and r^4 over the mixture matches σ_s^2 and the desired fourth moment $\mu_4 = (\tau_s + 3)\,\sigma_s^4$. Solving

$$w_1 r_1^2 + w_2 r_2^2 = \sigma_s^2, \qquad w_1 r_1^4 + w_2 r_2^4 = \mu_4$$

yields a unique non-negative solution for r_1, r_2 when $w_1 \;/\; w_2$ (we set $w_1 = w_2 = 1/2$ for simplicity and adjust radii accordingly). Because the directional vectors come from a 3-design, the composite ensemble with $M = 2M_0$ particles (two per direction) reproduces the target marginal distribution exactly in every projection and has isotropic covariance. Skewness is zero by symmetry. This initialization introduces **no** systematic error at birth; the empirical cumulants equal the true cumulants of the source distribution up to order four for all diagonal and off-diagonal quantities.

1.5 Propagation Through Layers

Linear layer. The weight matrix $W \in \mathbb{R}^{n \times n}$ acts identically on all particles of all sources: $\tilde{v}_m^{(s)} = W v_m^{(s)}$. The total pre-activation cloud before the non-linearity is the sum of the transformed particles plus the bias b.

ReLU non-linearity. The elementwise ReLU is applied to each particle of the combined cloud $\{z^{(p)}\}_{p=1}^{K}$ individually: $h_i^{(p)} = \max(0, z_i^{(p)})$. The post-ReLU cloud serves as input to the next

layer. We do **not** attempt to recluster particles back to individual sources; the entire cloud is treated as a single ensemble. This avoids the exponential fragmentation of particles across sources and keeps the particle count fixed at $K = kM$ throughout all layers.

1.6 Channel Extraction

At any pre-activation layer, the K particles $\{z^{(p)}\}$ deliver:

- **Mean:** $\hat{\mu}_i = \frac{1}{K}\sum_p z_i^{(p)}$.
- **Covariance:** $\hat{C}_{ij} = \frac{1}{K}\sum_p (z_i^{(p)} - \hat{\mu}_i)(z_j^{(p)} - \hat{\mu}_j)$.
- **Diagonal third cumulant:** $\hat{d}_3(i) = \frac{1}{K}\sum_p (z_i^{(p)} - \hat{\mu}_i)^3$.
- **Off-diagonal third cumulant Y_{21}:**

$$\hat{\kappa}_{2,1}(i,j) = \frac{K}{(K-1)(K-2)} \sum_p (z_i^{(p)} - \hat{\mu}_i)^2 (z_j^{(p)} - \hat{\mu}_j), \quad i \;/\; j. \qquad =$$

- **Diagonal fourth cumulant:** $\hat{K}_{4d}(i) = \frac{1}{K}\sum_p (z_i^{(p)} - \hat{\mu}_i)^4 - 3\hat{C}_{ii}^2$.

All quantities are computed from the same particle set. The computational bottleneck is Y_{21} at $O(Kn^2)$.

1.7 Integration with the Chain and Validation Protocol

The particle ensemble can replace the dense legs of any subset of sources. In the proposed scheme it replaces **all** sources from age 1 onward. The pre-activation mean, covariance, Y_{21}, and kurtosis diagonal are computed from the cloud and fed to the next layer's moment propagation, exactly as the teacher arrays are used in a one-step teacher-forced test.

1.7.1 Stage 1 Screen (Channel-by-Channel Check)

Using stored teacher arrays (true pre-activation mean, C, Y_{21}, K_{4d} at layers 3/7/11/14/15/16, obtained from a 60M-sample Monte Carlo), we evaluate two modes:

- **Teacher-forced one-step:** At layer ℓ, initialise the cloud with the true source distributions, propagate through W and ReLU, and compare the extracted channels to the teacher.
- **Free-run from L1:** Start from exact Gaussian source particles at L1, propagate layer by layer, and compare at the same deep layers.

The free-run is mandatory; a one-step test alone can hide missing memory errors (previous source-free schemes passed sanity checks but later diverged by 36–40% on Y_{21}).

The gate: on every channel (variance diagonal, d_3, K_{4d}, off-diagonal Y_{21}) the relative error must be $\leq 1.5\times$ the chain's error on that channel at the same layer. The chain's reference off-diagonal Y_{21} Frobenius errors are 4.6% (L7), 5.7% (L11), 6.2% (L14), 6.6% (L15). The scheme is killed if **any** channel exceeds $1.5\times$ the chain's error at two or more deep layers, or if the error growth rate with depth is steeper than the chain's.

1.7.2 Stage 2 Decisive (Final-Layer Mean MSE)

The particle scheme is integrated end-to-end (L1–L16). The output mean squared error (MSE) against the 10^9-sample truth is measured. The combined score is $\mathrm{MSE} \times \mathrm{cost}$. To be competitive, the score must be $\leq 0.44\times$ the chain's score, i.e. one of:

- $\mathrm{MSE} \leq 1.07\times$ chain at ≤ 102 u,
- $\mathrm{MSE} \leq 0.95\times$ chain at 115 u,
- $\mathrm{MSE} \leq 0.84\times$ chain at 125 u.

A prototype qualifies as a plausible path if it attains score $\leq 0.8\times$ chain (e.g. $\mathrm{MSE} \leq 1.6\times$ at 125 u) and the specific missing term responsible for the gap can be named. For the particle scheme the principal missing term is the cross-source Y_{21} correction:

$$\Delta Y_{21}(i,j) = \sum_{s \neq s'} \mathbb{E}\big[z_i^{(s)} z_i^{(s')} (z_j^{(s)} + z_j^{(s')})\big],$$

which arises because the particle cloud mixes sources after ReLU and does not preserve per-source identities. A hybrid strategy (dense legs for young sources, particles for old) can recover this term.

1.8 ONE Falsifier on Stored Data

The single decisive falsifier is the **free-run off-diagonal Y_{21} relative Frobenius error at L15**. Using the teacher Y_{21} array stored for L15, compute the particle scheme's Y_{21} from the K particles after 15 layers of unassisted propagation from L1. If the relative Frobenius norm of the difference exceeds 9.9% (i.e. $1.5\times$ the chain's 6.6% reference), the scheme fails immediately. Additionally, a monotonic growth pattern from L7 ($> 6.9\%$), L11 ($> 8.6\%$), L14 ($> 9.3\%$) with slope steeper than the chain's confirms failure of Stage 1 without requiring the full end-to-end MSE.

2. Per-Layer State, Cost, and Integration with Deterministic Particles

2.1 Motivation: Overcoming the Mixed-Rank Bottleneck

The chain currently maintains a heterogeneous state for the k sources that survive up to a given pre-activation layer. Young sources (ages 1–4) keep their full-rank $n \times n$ direct-space legs (P and A matrices, $n = 1024$), ages 5–7 share a rank-384 basis, and older sources are embedded in a nested sub-basis of rank 224. This age-dependent compression fails precisely where it matters most: the Hadamard products needed to form the off-diagonal third-moment channel $D_{21} \equiv \kappa_3(z_i, z_i, z_j)$ remain at full rank. Removing sources older than ten layers already inflicts a 16% error on D_{21} at layer 15, revealing that low-rank truncation discards cross-source information essential for deep layers.

We replace this entire hierarchy with a single, parsimonious state: for each of the k active sources at layer ℓ we store M deterministic particles – column vectors of length n – that serve as a compact approximate representation of the source's activation distribution. The total carried state is kMn scalars, orders of magnitude below n^3 when $M \ll n$ and $k \leq 14$. The particles are not random draws but a carefully constructed set that matches the mean, variance, skewness (d_3), diagonal excess kurtosis (K_{4d}), and the pairwise products required for D_{21}. Because each particle resides in the full n-dimensional space, no low-rank truncation is ever imposed on mixed moments; cross-source interactions that the mixed-rank chain disfigures are preserved naturally.

2.2 Construction and Content of the Particle Set

At the output of the ReLU non-linearity of layer ℓ, source s is represented by the matrix $\mathbf{X}_s \in \mathbb{R}^{n \times M}$, whose columns $\mathbf{x}_s^{(m)}$, $m = 1, \ldots, M$ are the particles. The set is deterministic and moment-matched: we select a quadrature rule or an extension of the unscented transform that not only matches the source' s mean and covariance (as the standard unscented transform does) but also aligns its marginal skewness and diagonal kurtosis to the target values prescribed by the analytic propagator. The same point set is used to estimate cross-source outer products for Y_{21}, the uncentered third-moment tensor that feeds D_{21}. No per-neuron basis change, age-dependent policy, or rank truncation is ever required.

2.3 Per-Layer State Evolution

The state evolves through each layer via three simple steps:

1. **Linear transform:** Every particle is multiplied by the weight matrix $\mathbf{W}^{(\ell+1)} \in \mathbb{R}^{n \times n}$, yielding pre-activation particles $\mathbf{z}_s^{(m)} = \mathbf{W}^{(\ell+1)} \mathbf{x}_s^{(m)}$.
2. **Element-wise activation:** The ReLU (or any subsequent activation) is applied individually: $\mathbf{a}_s^{(m)} = \max(0, \mathbf{z}_s^{(m)})$.
3. **Aggregation of summary channels:** From the post-activation particles we compute the quantities needed by the next layer' s analytic machinery. For each source we obtain the per-neuron mean $\boldsymbol{\mu}_s$, variance $\boldsymbol{\sigma}_s^2$, skewness $d_{3,s} = \kappa_3(z_i, z_i, z_i)$, and diagonal kurtosis $K_{4d,s}$. The off-diagonal Y_{21} matrix is assembled via sample moments; for example,

$$Y_{21}(i,j) = \frac{1}{M} \sum_{m=1}^{M} \big(z_{s,i}^{(m)}\big)^2 z_{s,j}^{(m)} - \boldsymbol{\mu}_{s,i}^2 \, \boldsymbol{\mu}_{s,j} - 2 \, \boldsymbol{\mu}_{s,i} \, \mathrm{Cov}_s(i,j) - \cdots ,$$

with additional central-moment corrections that remove finite-sample bias and ensure consistency with the independently estimated mean and covariance.

When new sources appear as depth increases (the chain' s counts grow from 6 to 14), they are simply allocated their own M initial particles drawn from the input distribution. No re-compression or basis realignment is necessary.

2.4 Cost Analysis in n^3-Equivalent Units

A single matrix-vector product costs n^2 multiply-add operations; propagating kM particles through $\mathbf{W}$ therefore consumes $kM\,n^2$ floating-point operations. Expressed in the natural unit of one full $n \times n$ matrix multiplication (a "square product"), the equivalent number of square products per layer is kM/n. With $n = 1024$, $M = 16$, and $k \le 14$,

$$\frac{kM}{n} \le \frac{14 \times 16}{1024} = \frac{224}{1024} \approx 0.22 \text{ square products.}$$

The aggregation step adds a second block of $O(kM\,n^2)$ work because Y_{21} can be built by accumulating rank-one updates: for each particle we form the outer product $\left(\mathbf{z}^{(m)} \odot \mathbf{z}^{(m)}\right)\mathbf{z}^{(m)\top}$, an $O(n^2)$ operation. The total for both phases therefore costs at most $2kM\,n^2$, or $2kM/n \approx 0.44$ square products per layer. Additional $O(kMn)$ scalar moment calculations are negligible.

The chain's allocated total budget is ≤ 125 u, where "u" is calibrated such that a full forward pass using fifteen n^3 square products per layer would still be permissible (i.e., 125 u corresponds roughly to ≤ 15 square products per layer across the network). The particle scheme consumes less than half a square product per layer, leaving over 99% of the budget untouched. Memory is likewise minimal: the state consists of $kMn \approx 2.3 \times 10^5$ floats, fitting comfortably in on-chip SRAM.

2.5 Seamless Integration into the Chain

Replacing the per-source dense legs and mixed-rank bases demands no architectural change. Internally, the chain retains its teacher-compatible interface: functions that return the mean vector, covariance matrix, Y_{21} tensor, and K_{4d} diagonal keep identical signatures. The new implementation simply populates these arrays from the particle moments. The downstream analytic machinery – the same that computes next-layer mean and variance under ReLU, or the off-diagonal D_{21} correction – continues to operate unchanged.

During Stage 1 verification, the scheme can be run either in "teacher-force" mode (true layer-ℓ inputs are used to compute layer-$\ell+1$ channels) or free-running from layer 1 onward. The particle-derived channels – variance, d_3, K_{4d}, and off-diagonal D_{21} – will be compared against 60M-sample teacher arrays at layers $3, 7, 11, 14, 15, 16$. The acceptance gate is stringent: no channel error may exceed the chain's current error (e.g., 4.6–6.6% for D_{21} off-diagonal), and no channel may be $\ge 1.5\times$ worse at two or more deep layers. Stage 2 integrates the scheme end-to-end and measures the final-layer mean MSE against the 10^9-sample truth, targeting a combined score $\mathrm{MSE} \times \mathrm{cost} \le 0.44\times$chain. With $M = 16$ the particle approximation is expected to be a plausible path, achieving at worst $\mathrm{MSE} \le 1.6\times$chain at 125u and leaving a clear residual (insufficient moment matching for deep skew–kurtosis interactions) that can be closed by augmenting the particle set with a modest number of additional points.

3. Falsifier and Two-Stage Evaluation Plan

3.1 Deterministic Nyström-Tail Compression (dNTC)

The proposed scheme avoids carrying the full per-source dense (P_s, A_s \in \mathbb{R}^{n \times n}) (n=1024) and the full third-order tensor (\kappa_3). It replaces each old source with a deterministic low-rank-plus-diagonal approximation built from a shared set of (m=120) landmark coordinates (\mathcal{J} \subset {1,\dots,n}). The state propagated per source is:

- a landmark cross-matrix (C_s^{\text{(land)}} \in \mathbb{R}^{m \times n}),
- a diagonal correction vector (\delta_s \in \mathbb{R}^n),
- a small core tensor (\mathcal{T}_s^{\text{(core)}} \in \mathbb{R}^{m \times m \times m}) for third-order contributions,
- a length-(n) vector (\kappa_{s,4}^{\text{(diag)}}) for the diagonal of the fourth cumulant.

These objects are updated recursively using the previous layer’ s activation statistics and weight structure, without ever forming full (n \times n) or (n \times n \times n) arrays. The approximate covariance is assembled as [\hat{C} \approx D + \sum_s C_s^{\text{(land)},T} \bigl(C_s^{\text{(land)}} C_s^{\text{(land)},T}\bigr)^{\dagger} C_s^{\text{(land)}},] where (D = \operatorname{diag}!\bigl(\sum_s \delta_s\bigr)). All operations reduce to matrix-vector products with the (m \times n) landmark matrices and inversion of small (m \times m) blocks. With (m=120) the per-layer update costs at most 15 square products (measured in (n^2) multiply-add equivalents), well within the ⩽125u budget (⩽15 square products per layer).

The **single falsifier** is a free-run channel-wise error gate: the dNTC state is propagated from layer1 onward, the per-neuron channels (variance, (d_3 = \kappa_3(z_i,z_i,z_i)), (\kappa_4) diagonal, off-diagonal (C), and D21) are extracted at layers 7, 11, 14, and 15, and compared against the teacher truth arrays.

3.2 Stage1 — Channel-Error Gate on Stored Arrays

We run dNTC in free-running mode, without access to intermediate teacher statistics, using only the true input distribution. At each gate layer the following channels are compared with the 60M-sample Monte-Carlo teacher arrays (pre-activation and post-ReLU at layers 3/7/11/14/15):

- per-neuron variance ((\operatorname{diag}(C))), split into diagonal relative error and off-diagonal Frobenius norm error via D21;
- diagonal third central moment (d_3);
- diagonal fourth cumulant (\kappa_4);
- off-diagonal covariance D21 relative Frobenius error (|\text{off}(C)|_F / |C|_F).

All errors are relative and computed per layer. The chain’ s own errors on these channels are measured on the same teacher arrays as baseline; the known chain D21 errors are:

Layer	Chain D21 error (%)
L7	4.6

Layer	Chain D21 error (%)
L11	5.7
L14	6.2
L15	6.6

The chain’s errors for variance, (d_3), and (\kappa_4) diagonal are computed analogously and serve as gate references.

Kill criteria. dNTC is rejected if **any** of the following occurs:

- The error on **any** channel (variance, (d_3), (\kappa_4) diagonal, off-diagonal (C), D21) exceeds **1.5 ×** the chain’s error on that channel at **two or more** of the deep layers {11, 14, 15}.
- The error growth with depth is steeper than the chain’s; specifically, the slope of log-error vs. layer index (linear fit over L7–L15) for dNTC exceeds the chain’s slope by more than 20%.

A pass through this gate marks the scheme as a plausible candidate. Failure immediately kills the prototype; the layer and channel where the breakdown occurs is identified and the missing compression term is flagged.

3.3 Stage 2 — Final-Layer Mean MSE Integration (Decisive)

The dNTC state is integrated into the full chain: sources aged 1–4 are kept in the exact dense representation, while sources aged $\geqslant 5$ are handled by the dNTC compressed state. The evaluation runs on the same 100 MLP configurations (architecture, activation, initialisation) from the competition dataset.

We measure the final-layer (layer 16) empirical mean squared error (MSE) against a 1e9-sample Monte-Carlo truth for each MLP, paired with the baseline chain’s MSE from an identical run. The aggregated score is $[\text{Score} = \text{MSE}_{\text{dNTC}} \times \text{cost},]$ where cost is in units “u” (1 u ≈ one (n^2) multiply-add). The top-3 viability thresholds are:

- Score $\leqslant 0.44 \times$ chain, achievable via:
  - MSE $\leqslant 1.07 \times$ chain at cost $\leqslant 102$ u, or
  - MSE $\leqslant 0.95 \times$ chain at cost $\leqslant 115$ u, or
  - MSE $\leqslant 0.84 \times$ chain at cost $\leqslant 125$ u.

A prototype that does not meet the top-3 threshold but stays within Score $\leqslant 0.8 \times$ chain qualifies as a **plausible path** and is kept for further development. In this case the report must name the specific missing term (e.g., “off-diagonal (\kappa_3) contribution from sources aged $\geqslant 8$”) that prevents closing the gap. Any scheme with Score > $0.8 \times$ chain is considered dead.

The relative error metric is the normalised L2 distance ($| \mu_{\text{dNTC}} - \mu_{\text{true}} |^2 / | \mu_{\text{chain}} - \mu_{\text{true}} |^2$), averaged over the 100 models.

Cost is profiled by counting (n^2) products inside the critical loop; the lowest cost among ten realisations of the random landmark set (\mathcal{J}) is used to ensure a fair comparison.